

FBISE CHEMISTRY

SSC-I / CLASS 9

New Model Paper 2026 - Complete Solved Paper

60 MCQs with reasoning | Complete Section B | Complete Section C
All OR options attempted

PREPARED BY

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Student Preparation Edition

Based on the uploaded Federal Board SSC-I Chemistry Model Question Paper
National Curriculum 2022-2023 (Inclusive Scheme of Studies 2024)

How to Use This Solution

- First attempt the model paper yourself under timed conditions, then compare with these solutions.
- For MCQs, focus on the reason, not only the option. The same concept can be asked with different wording.
- For Section B and C, copy the answer structure: definition/principle, working or equation, and a clear conclusion.
- In numerical questions, always write the formula, substitute values with units, and state the final answer.
- In chemical equations, balance atoms on both sides and include conditions where they are important.

Paper Pattern

Section	Marks	What to Prepare
A	30	60 MCQs × 0.5 mark
B	18	6 short questions × 3 marks
C	12	2 long questions × 6 marks

Important Note

This solution is an independently prepared educational resource by Fakhar STEM Sphere. It is designed for exam preparation and follows the questions in the supplied FBISE model paper.

SECTION A - MCQs

Quick Answer Key

Q	Ans	Q	Ans	Q	Ans
1	C	21	D	41	B
2	C	22	C	42	B
3	B	23	A	43	A
4	B	24	C	44	B
5	D	25	D	45	C
6	A	26	A	46	C
7	B	27	B	47	C
8	A	28	D	48	C
9	C	29	A	49	B
10	C	30	A	50	B
11	B	31	A	51	A
12	B	32	C	52	C
13	C	33	D	53	C
14	C	34	A	54	A
15	B	35	C	55	B
16	C	36	D	56	C
17	B	37	B	57	B
18	B	38	B	58	B
19	D	39	C	59	B
20	C	40	D	60	B

Detailed MCQ Solutions

MCQ 1. Diamond is used in cutting tools. The reason for this use is:

Correct option: C

Diamond has a giant three-dimensional covalent network. Each carbon atom is strongly covalently bonded to four other carbon atoms.

Because strong covalent bonds extend throughout the crystal, a very large amount of energy is required to deform or break the structure. This makes diamond extremely hard and suitable for cutting tools.

Option A describes graphite more closely: graphite has layers with weak forces between the layers.

MCQ 2. Which of the following is a compound? Air, iron, carbon dioxide, brass.

Correct option: C

A compound contains two or more different elements chemically combined in a fixed ratio. Carbon dioxide contains carbon and oxygen chemically combined as CO_2 .

Air is a mixture of gases, iron is an element, and brass is an alloy (mixture). Therefore carbon dioxide is the compound.

MCQ 3. Which combination is correctly matched?

Correct option: B

Milk is a colloid: tiny fat droplets are dispersed through water and do not settle quickly.

Sugar + water forms a true solution, sand + water forms a suspension, and air is a homogeneous gaseous mixture.

MCQ 4. The shells around the nucleus are also called:

Correct option: B

Electron shells correspond to definite energy levels around the nucleus. Therefore the shells are also called energy levels.

An orbital is a region of high probability of finding an electron and is not the same as a complete shell.

MCQ 5. Identify the numbers of electrons, protons and neutrons in $^{58}_{26}\text{X}$.

Correct option: D

The atomic number 26 gives the number of protons. Because the atom is neutral, electrons = protons = 26.

Neutrons = mass number - atomic number = 58 - 26 = 32.

Electrons = 26, Protons = 26, Neutrons = 32

MCQ 6. An element has isotopes of mass 24 (90%) and mass 25 (10%). Find its relative atomic mass.

Correct option: A

Relative atomic mass is the weighted average of the isotope masses.

$$A_r = [(24 \times 90) + (25 \times 10)] / 100 = 24.1$$

MCQ 7. What is the formula of aluminium oxide formed from Al^{3+} and O^{2-} ?

Correct option: B

The total positive and negative charges must cancel. The LCM of 3 and 2 is 6, so two Al^{3+} ions give +6 and three O^{2-} ions give -6.

Therefore the neutral formula is Al_2O_3 .



MCQ 8. Which change is an example of oxidation?

Correct option: A

Oxidation is loss of electrons or an increase in oxidation number. A chloride ion must lose an electron to become a neutral chlorine atom.

The other listed ionic changes involve gain of electrons and are reductions.



MCQ 9. In which change is nitrogen reduced?

Correct option: C

In N_2 , nitrogen has oxidation number 0. In NH_3 , nitrogen has oxidation number -3.

The oxidation number decreases from 0 to -3, so nitrogen is reduced.



MCQ 10. X and Y are diatomic elements. X is less reactive than Y. Which pair fits?

Correct option: C

For halogens, reactivity decreases down Group VII/17: $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$.

Iodine is less reactive than bromine, so X = iodine and Y = bromine.

MCQ 11. Which pair of properties is correct for a typical transition element?

Correct option: B

Transition elements typically have high density and can show variable oxidation states. They also commonly form coloured compounds and act as catalysts.

Options that include low density or low melting point are not typical.

MCQ 12. Which two displayed compounds are members of the same homologous series?

Correct option: B

Compound 1 is methanol (CH_3OH) and compound 3 is ethanol ($\text{C}_2\text{H}_5\text{OH}$). Both contain the hydroxyl ($-\text{OH}$) functional group and belong to the alcohol homologous series.

Compound 2 is a carboxylic acid and compound 4 is an ether.

MCQ 13. Which displayed structure is not an isomer of $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_2\text{-CH}_3$?

Correct option: C

An isomer must have the same molecular formula but a different structural arrangement. The given compound is n-pentane, C_5H_{12} .

Option C is simply the same straight five-carbon chain drawn with the last CH_3 group turned downward; connectivity has not changed. Therefore it is not an isomer - it is the same compound.

MCQ 14. Which two elements react together to form an ionic compound? W: 2,4; X: 2,8; Y: 2,8,1; Z: 2,8,7.

Correct option: C

Y has one valence electron and tends to lose it to form Y^+ . Z has seven valence electrons and tends to gain one to form Z^- .

Electron transfer from Y to Z produces oppositely charged ions, so Y and Z form an ionic compound.

MCQ 15. For $^{52}_{23}X$ and $^{52}_{24}Y$, which statement is correct?

Correct option: B

The proton (atomic) number identifies an element. The first atom has 23 protons and the second has 24 protons. Because their proton numbers are different, they are different elements, even though both have mass number 52.

Checkpoint: Revise the concepts above before moving to the next group of MCQs.

MCQ 16. The atoms $^{64}_{29}Cu$ and $^{65}_{30}Zn$ have the same:

Correct option: C

For Cu: neutrons = $64 - 29 = 35$. For Zn: neutrons = $65 - 30 = 35$.

Therefore they have the same number of neutrons.

MCQ 17. Why does molten sodium chloride conduct electricity?

Correct option: B

In solid NaCl, Na^+ and Cl^- ions are fixed in a crystal lattice and cannot move through the solid.

When NaCl melts, the lattice breaks down and the ions become mobile. These moving ions carry electric charge through the liquid.

MCQ 18. Which change is endothermic?

Correct option: B

The change $H_2O(g) \rightarrow 2H(g) + O(g)$ requires the O-H bonds in water to be broken.

Bond breaking absorbs energy, so this process is endothermic.

MCQ 19. Why is $C_2H_4 + 3O_2 \rightarrow 2CO_2 + 2H_2O$ exothermic?

Correct option: D

Energy is absorbed to break reactant bonds and released when product bonds form.

An exothermic reaction releases energy overall, meaning the energy required to break bonds is less than the energy released when new bonds form.

MCQ 20. What does the given energy profile show about the reaction and ΔH ?

Correct option: C

The diagram places the products at a lower energy level than the reactants. Therefore energy has been released.

For an exothermic reaction, $\Delta H = H(\text{products}) - H(\text{reactants})$ is negative.

Exothermic reaction: $\Delta H < 0$

MCQ 21. For $xCu + yHNO_3 \rightarrow xCu(NO_3)_2 + 4H_2O + 2NO$, find x and y.

Correct option: D

Four water molecules contain 8 hydrogen atoms, so $y = 8$. The 8 nitrate acid molecules supply 8 nitrogen atoms. On the product side there are $2x$ nitrogen atoms in $Cu(NO_3)_2$ plus 2 nitrogen atoms in $2NO$. Thus $2x + 2 = 8$, so $x = 3$.



MCQ 22. What is the empirical formula of P_4O_{10} ?

Correct option: C

An empirical formula gives the simplest whole-number ratio of atoms. Divide both subscripts by their highest common factor, 2.



MCQ 23. What is the mass of 3.0×10^{23} molecules of CO_2 ?

Correct option: A

First convert particles to moles using Avogadro's constant. 3.0×10^{23} molecules is approximately 0.50 mol.

The molar mass of CO_2 is $12 + 2(16) = 44 \text{ g mol}^{-1}$.

$$n = (3.0 \times 10^{23}) / (6.02 \times 10^{23}) \approx 0.50 \text{ mol}$$

$$m = nM = 0.50 \times 44 \approx 22 \text{ g}$$

MCQ 24. Hydrogen and chlorine react: one molecule of each forms two molecules of hydrogen chloride. Choose the equation.

Correct option: C

Elemental hydrogen and chlorine are diatomic: H_2 and Cl_2 . One H_2 molecule plus one Cl_2 molecule gives two HCl molecules.



MCQ 25. What is the total number of atoms in 2.00 moles of nickel?

Correct option: D

One mole contains 6.02×10^{23} atoms. Multiply by 2.00 moles.

$$N = 2.00 \times 6.02 \times 10^{23} = 1.204 \times 10^{24} \approx 1.2 \times 10^{24} \text{ atoms}$$

MCQ 26. Which statement about alkanes is correct?

Correct option: A

Alkanes are saturated hydrocarbons with general formula $\text{C}_n\text{H}_{2n+2}$. They mainly undergo substitution rather than addition reactions.

As carbon-chain length increases, molecules become larger and intermolecular attractions become stronger, so boiling points generally increase.

MCQ 27. Which group contains compounds all from the same homologous series?

Correct option: B

CH_4 , C_2H_6 and C_3H_8 all satisfy the alkane general formula $\text{C}_n\text{H}_{2n+2}$.

Therefore all three belong to the same homologous series: the alkanes.

MCQ 28. Butane and methylpropane are isomers. Which formula is different for the two?

Correct option: D

Both have the same molecular formula, C_4H_{10} , and therefore the same empirical formula. They also belong to the same alkane homologous series.

Isomers differ in how atoms are connected, so their structural formulae are different.

MCQ 29. Which gas cannot be removed from petrol-engine exhaust by a catalytic converter?

Correct option: A

A catalytic converter oxidises carbon monoxide and unburnt hydrocarbons and reduces nitrogen oxides.

Carbon dioxide is not removed by the converter; it is actually formed when carbon monoxide and hydrocarbons are oxidised.

MCQ 30. Which noble gas is present in the largest percentage by volume in air?

Correct option: A

Argon is the most abundant noble gas in Earth's atmosphere (about 0.93% by volume), far more abundant than neon, helium or krypton.

Checkpoint: Revise the concepts above before moving to the next group of MCQs.

MCQ 31. Ammonia gas is produced when solid ammonium chloride is heated with:

Correct option: A

Ammonium salts release ammonia when heated with a strong alkali. Calcium hydroxide is the alkali among the options.



MCQ 32. Which shown reaction is not a typical reaction of a dilute acid?

Correct option: C

Typical school-level reactions of dilute acids include neutralisation with alkalis/bases and reaction with carbonates to give CO_2 .

Option C is an acid + soluble salt precipitation reaction: $\text{Pb}(\text{NO}_3)_2 + \text{H}_2\text{SO}_4$ gives insoluble PbSO_4 . It is therefore the non-typical option in this set.

MCQ 33. What is the ionic equation for zinc reacting with aqueous copper(II) sulfate?

Correct option: D

The molecular equation is $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$. Sulfate ions remain unchanged and are spectator ions. Cancel SO_4^{2-} from both sides to obtain the net ionic equation.



MCQ 34. Blue $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ turns to white CuSO_4 when heated. Which way did equilibrium shift?

Correct option: A

Heating removes water of crystallisation, converting hydrated blue copper(II) sulfate into white anhydrous copper(II) sulfate.

The dehydration direction absorbs heat, so heating favours the forward, endothermic reaction.



MCQ 35. Which series includes both oxidation and reduction?

Correct option: C

In $\text{N}_2 \rightarrow \text{NH}_3$, nitrogen changes from oxidation number 0 to -3, so it is reduced.

In $\text{NH}_3 \rightarrow \text{NO}$, nitrogen changes from -3 to +2, so it is oxidised. Hence option C contains both processes.

MCQ 36. A colourless gas leaves KI colourless, turns acidified dichromate orange to green and acidified permanganate purple to colourless. What type is it?

Correct option: D

Acidified dichromate and permanganate change colour when they are reduced. The gas causes both oxidising agents to be reduced.

Therefore the gas itself acts as a reducing agent.

MCQ 37. Enzymes involved in digestion are:

Correct option: B

Enzymes are biological catalysts. Most enzymes are proteins with specific three-dimensional structures suited to their substrates.

MCQ 38. A solution turns litmus blue. What is its nature and possible pH?

Correct option: B

A basic solution turns red litmus paper blue. Bases have pH values greater than 7.

Among the options, basic with pH 10 is correct.

MCQ 39. Why do elements in the same group show similar chemical behaviour?

Correct option: C

Elements in the same group have the same number of valence electrons. Chemical reactions mainly involve these outer-shell electrons.

Therefore they tend to form similar ions and compounds and show similar chemical behaviour.

MCQ 40. Which element has the largest atomic size: F, O, Cl or Br?

Correct option: D

Atomic size increases down a group because extra electron shells are added. Bromine is below chlorine and fluorine and therefore has the largest radius among the listed elements.

Checkpoint: Revise the concepts above before moving to the next group of MCQs.

MCQ 41. Scientists studying how enzymes function in the human body are working in which field?

Correct option: B

Biochemistry studies the substances and chemical processes of living organisms. Enzymes and their action in the body are therefore studied in biochemistry.

MCQ 42. The development of new medicines to treat diseases belongs to:

Correct option: B

Among the given options, organic chemistry is the best match because many medicines are carbon-based compounds whose structures, synthesis and reactions fall within organic chemistry.

MCQ 43. The use of radioactive isotopes in cancer treatment is related to:

Correct option: A

Radioactive isotopes undergo nuclear changes and their medical applications are studied under nuclear chemistry.

MCQ 44. Why are different laboratories needed in the research centre?

Correct option: B

Different branches of chemistry address different kinds of substances, techniques, hazards and applications. Separate specialist laboratories therefore support different research problems and equipment needs.

MCQ 45. What is the process when liquid wax turns into a solid?

Correct option: C

The change from liquid to solid is called freezing or solidification. Therefore the correct option is freezing.

MCQ 46. Why is the freezing of wax a physical change?

Correct option: C

No new substance is formed. The wax has the same chemical composition before and after freezing, and the change can be reversed by melting.

MCQ 47. What happens to wax particles during freezing?

Correct option: C

As wax cools, particles lose kinetic energy and become more closely packed and ordered. Their chemical identity does not change.

MCQ 48. Which property remains unchanged when liquid wax freezes?

Correct option: C

The state and shape can change, and volume may change slightly, but the chemical composition remains unchanged because no chemical reaction occurs.

MCQ 49. An atom has 17 protons and 18 neutrons. What is its mass number?

Correct option: B

Mass number = number of protons + number of neutrons.

$$A = 17 + 18 = 35$$

MCQ 50. An element has 17 protons. Which group and period does it belong to?

Correct option: B

Atomic number 17 is chlorine with electronic configuration 2,8,7. Three occupied shells place it in Period 3; seven valence electrons place it in Group 17 (Group VII).

MCQ 51. What is the electronic configuration of the element with 17 electrons?

Correct option: A

Fill the shells: 2 in the first shell, 8 in the second, leaving 7 in the third.

$$\text{Electronic configuration} = 2, 8, 7$$

MCQ 52. What is the atomic number of the atom with 17 protons?

Correct option: C

Atomic number is defined as the number of protons in the nucleus. Therefore its atomic number is 17.

Checkpoint: Revise the concepts above before moving to the next group of MCQs.

MCQ 53. After methane, ethane, propane and butane, what is the next compound in the same homologous series?

Correct option: C

These are alkanes with general formula C_nH_{2n+2} . For $n = 5$, the formula is C_5H_{12} , called pentane.

MCQ 54. What is the common difference between successive members of a homologous series?

Correct option: A

Successive members of a homologous series differ by one $-CH_2-$ unit, which increases relative molecular mass by 14.

MCQ 55. Why do members of the same homologous series show similar chemical properties?

Correct option: B

Members have the same characteristic functional group (and the same general type of bonding), so they undergo similar chemical reactions.

MCQ 56. Which property shows a gradual change in a homologous series?

Correct option: C

Physical properties such as boiling point change gradually as molecular size increases. The functional group remains the same throughout the series.

MCQ 57. Which traffic-related gas is especially dangerous because it binds with haemoglobin?

Correct option: B

Carbon monoxide binds strongly with haemoglobin and prevents normal oxygen transport in the blood. Therefore CO is especially dangerous.

MCQ 58. Why is carbon monoxide toxic?

Correct option: B

CO combines with haemoglobin to form carboxyhaemoglobin. This greatly reduces the blood's ability to carry oxygen to tissues, which can cause headache, unconsciousness and death at high exposure.

MCQ 59. Which listed gas contributes most to global warming?

Correct option: B

Among CO , CO_2 , NO_2 and SO_2 , carbon dioxide is the principal greenhouse gas considered in this school-level context. It absorbs outgoing infrared radiation and contributes to the greenhouse effect.

MCQ 60. What is the main source of nitrogen oxides in cities?

Correct option: B

High-temperature combustion in motor-vehicle engines and other fossil-fuel burning produces nitrogen oxides (NO_x). Therefore traffic and fossil-fuel combustion are the major urban source among the choices.

SECTION B - Short Questions ($6 \times 3 = 18$ Marks)

All questions and all OR options are solved below. In the examination, write only the option you choose unless instructed otherwise.

Q1 - Group VII Elements and Trends

Group VII (modern Group 17) elements are fluorine, chlorine, bromine, iodine and astatine.

- Physical state down the group: fluorine and chlorine are gases, bromine is a liquid, while iodine and astatine are solids at room temperature.
- Thus the physical state changes generally from gas $\rightarrow$ liquid $\rightarrow$ solid as we move down the group.
- Density generally increases down the group because the atoms/molecules become heavier and more polarisable.

Full-marks form: Group VII elements are F, Cl, Br, I and At. Down the group, the physical state changes from gases to a liquid and then to solids, while density generally increases.

Q1 - OR: Compound vs Mixture

Compound	Mixture
Two or more elements are chemically combined.	Two or more substances are physically mixed.
Composition is fixed.	Composition may vary.
Has a definite chemical formula.	Has no fixed chemical formula.
Constituents lose their individual identity/properties and the compound has new properties.	Constituents retain their individual properties.
Components require chemical methods to separate.	Components can usually be separated by physical methods.

Examples: H_2O , NaCl and CO_2 are compounds; air, salt solution and brass are mixtures.

Q2 - Compare Reactivity of Metals and Non-metals Using Electron Behaviour

- Metals react mainly by losing valence electrons and forming positive ions: $\text{M} \rightarrow \text{M}^{n+} + n\text{e}^-$.
- Down a metallic group, atomic size and shielding increase. The outer electron is less strongly attracted to the nucleus, so it is lost more easily. Hence metallic reactivity generally increases down a group.
- Non-metals react mainly by gaining electrons and forming negative ions: $\text{X} + \text{e}^- \rightarrow \text{X}^-$.
- Down a non-metal group such as the halogens, size and shielding increase, so attraction for an incoming electron becomes weaker. Hence halogen reactivity decreases down the group: $\text{F} > \text{Cl} > \text{Br} > \text{I}$.

Q2 - OR: Electroplating to Prevent Corrosion

Electroplating is the process of depositing a thin layer of one metal on another object by electrolysis.

- Clean the object thoroughly so the coating can stick properly.
- Make the object to be plated the cathode (-).
- Use the coating metal as the anode (+).
- Use an electrolyte containing ions of the coating metal, and pass direct current through it.



The deposited metal layer forms a barrier that prevents oxygen and moisture from reaching the underlying metal, so corrosion is reduced. Electroplating may also improve appearance and hardness.

Q3 - Air Pollution Scenario

Most likely pollution condition: smog, especially smog containing fine particulate matter and nitrogen oxides.

- Health effect: fine particles and irritant gases enter the respiratory tract, causing coughing, eye/throat irritation, breathing difficulty and worsening of asthma or bronchitis.
- Visibility effect: suspended particles and droplets scatter/absorb light, producing haze and reducing visibility.
- Environmental effect: photochemical oxidants and particles can damage plants, reduce photosynthesis, and deteriorate materials and buildings.

Exam wording tip: If your textbook names a specific pollutant for this symptom set, write that exact term first, then explain respiratory irritation and reduced visibility.

Q3 - OR: Three Main Classes of Biomolecules in Food

Biomolecule	Main role	Common food sources	Typical adult intake guidance*
Carbohydrates	Main energy source	Bread, rice, wheat, cereals, potatoes, fruits	About 45-65% of daily energy
Proteins	Growth, repair, enzymes	Meat, fish, eggs, milk, pulses, beans, lentils	About 0.8 g per kg body mass per day
Lipids / fats	Concentrated energy, insulation, vitamins	Oils, nuts, seeds, butter/ghee, dairy	About 20-35% of daily energy

*The model paper does not itself provide fixed numerical dietary values. The figures above are commonly used general nutrition guidance. If the prescribed FBISE textbook gives specific daily amounts, use those exact textbook values in the examination.

Q4 - Reactions of Hydrochloric Acid

(a) With sodium metal



A salt (sodium chloride) and hydrogen gas are formed.

(b) With sodium carbonate



Effervescence occurs because carbon dioxide gas is evolved.

(c) With sodium hydroxide



This is a neutralisation reaction: acid + base → salt + water.

Q4 - OR: Distilled Water, Tap Water and Activated Carbon

- Distilled water is nearly pure H_2O obtained by vaporising water and condensing the vapour. It contains very little dissolved mineral matter.
- Tap water normally contains dissolved mineral ions, dissolved gases and small amounts of water-treatment chemicals such as chlorine.
- Activated carbon has a very large porous surface. It adsorbs many dissolved organic substances and helps remove unpleasant taste, odour and residual chlorine.

Activated carbon therefore improves water quality beyond simply removing visible particles. However, it does not by itself guarantee removal of all dissolved salts or microorganisms.

Q5 - Identify Functional Groups

Compound	Functional group	Name of functional group
CH_3COOH	-COOH	Carboxyl group
CH_3CHO	-CHO	Aldehyde group
$\text{CH}_3\text{CH}_2\text{OH}$	-OH	Hydroxyl group

Q5 - OR: Reversible Reaction and Equilibrium

A reversible reaction can proceed in both the forward and reverse directions under suitable conditions.



- Dynamic equilibrium is reached when the rate of the forward reaction becomes equal to the rate of the reverse reaction.
- The system should be closed so reactants/products cannot escape, and suitable conditions such as temperature should be maintained.
- At equilibrium, both reactions continue, but the concentrations and macroscopic properties remain constant.

Important: equilibrium does not mean the reactions have stopped; it is dynamic.

Q6 - Isotopes and Chemical Properties

(a) Yes, the two atoms are isotopes of the same element.

$$\text{First atom: } A = 17 + 18 = 35 \rightarrow {}^{35}_{17}\text{Cl}$$

$$\text{Second atom: } A = 17 + 20 = 37 \rightarrow {}^{37}_{17}\text{Cl}$$

Both have the same proton number (17) but different neutron numbers and therefore different mass numbers. This is exactly the definition of isotopes.

(b) Their chemical properties are very similar because neutral atoms of both isotopes contain 17 electrons and have the same electronic configuration, 2,8,7. Chemical properties depend mainly on valence electrons. Their physical properties may differ slightly because their masses differ.

Q6 - OR: Define Chemistry and Explain States of Matter in Chemical Interactions

Chemistry is the branch of science that studies the composition, structure, properties and transformations of matter, together with the energy changes accompanying those transformations.

- Solids: particles are closely packed and mainly vibrate about fixed positions. Solid reactions often occur at surfaces or after heating, dissolving or melting.
- Liquids: particles are close but mobile, so substances can mix and dissolved particles can collide readily. Many reactions are carried out in solution.
- Gases: particles are far apart and move rapidly; reactions occur when particles collide with sufficient energy and suitable orientation.
- Plasma: an ionised state containing mobile ions and electrons. It conducts electricity and is highly energetic/reactive.

The state of matter changes particle spacing and mobility, which affects collision frequency and therefore how chemical reactions proceed.

SECTION C - Long Questions ($2 \times 6 = 12$ Marks)

Write long answers in a logical sequence. Use equations, labelled points and cause-and-effect explanations.

Q7 - Ionic Bonding from the Observed Properties

Given properties: hard crystalline solid, very high melting point, does not conduct electricity as a solid, but conducts when molten.

1. Type of bonding

These observations are characteristic of an ionic solid, so the bonding is ionic.

2. Formation of ionic bonding

Ionic bonding forms when electron(s) are transferred from one atom (usually a metal) to another atom (usually a non-metal). This creates positive and negative ions which attract each other electrostatically.



The electrostatic attraction between Na^+ and Cl^- is an ionic bond. In a solid, many ions form a giant three-dimensional ionic lattice.

3. High melting point

Strong electrostatic attractions act in all directions between oppositely charged ions in the giant lattice. A large amount of energy is needed to overcome these forces, so the melting point is high.

4. Electrical conductivity

- Solid state: ions are locked at fixed positions in the lattice. They cannot move and therefore cannot carry current.
- Molten state: the lattice has broken down and the ions are free to move. These mobile ions carry charge, so the melt conducts electricity.

Full-marks conclusion: The combination of high melting point, crystalline structure and conduction only when molten is strong evidence for ionic bonding and a giant ionic lattice.

Q7 - OR: Ethane - Reactivity, Reactions and Preparation

Ethane (C_2H_6) is a saturated hydrocarbon containing only C-C and C-H single covalent bonds.

Why ethane is generally unreactive

Ethane has no $\text{C}=\text{C}$ or $\text{C}\equiv\text{C}$ multiple bond and no highly reactive functional group. Its C-C and C-H sigma bonds are relatively strong, so it is comparatively unreactive under ordinary conditions.

Combustion



Complete combustion is exothermic and produces carbon dioxide and water.

Substitution with chlorine



A chlorine atom replaces a hydrogen atom, so this is a substitution reaction.

Preparation by hydrogenation of an alkyne



Preparation from an alkyl halide



A commonly taught alkyl-halide route is also the Wurtz reaction:

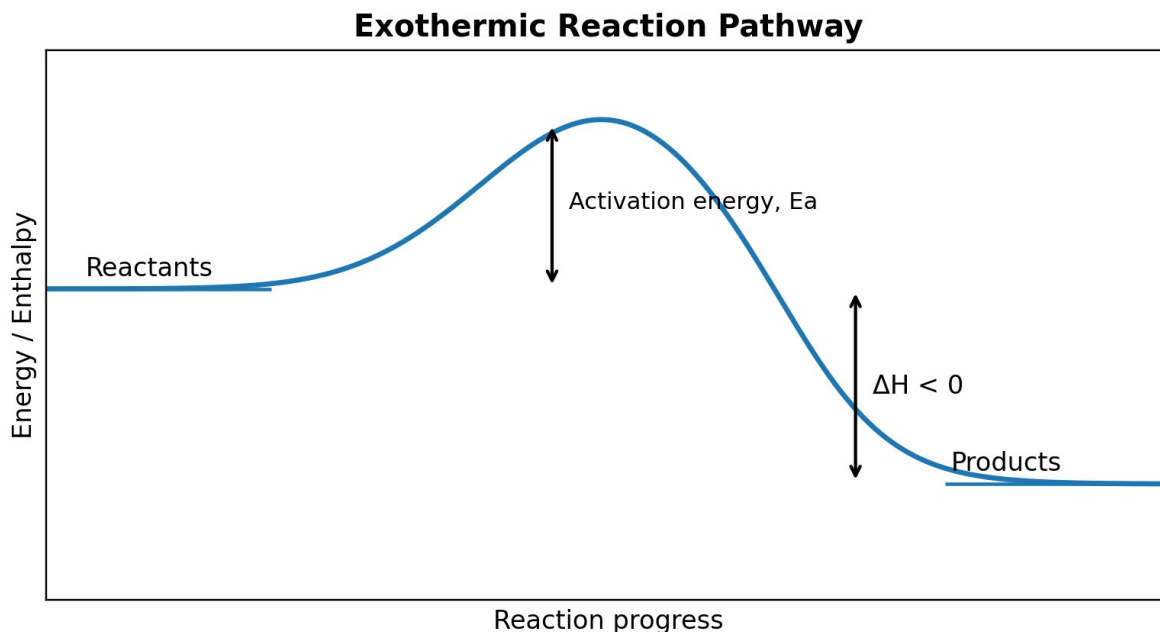


Note: The wording in the model paper asks for preparation from alkynes and alkyl halides. The equations above directly address those two sources.

Q8 - Reaction Pathway Diagrams

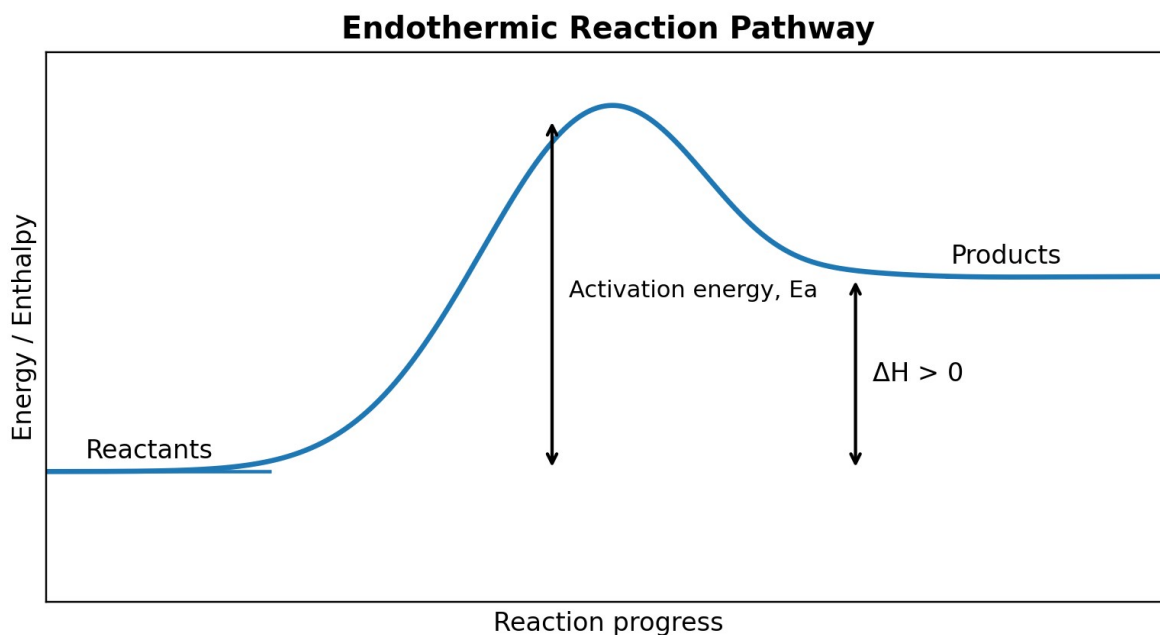
For every energy profile, label the vertical axis as energy/enthalpy and the horizontal axis as reaction progress. Show reactants, products, activation energy (E_a) and enthalpy change (ΔH).

Exothermic reaction



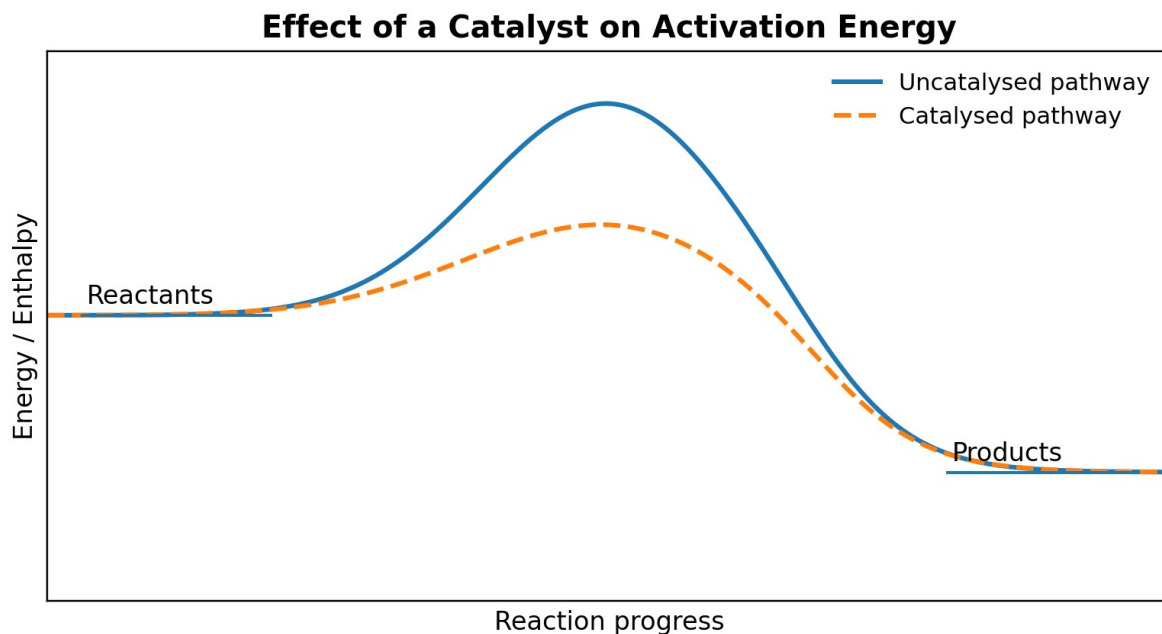
- Products are at lower energy than reactants.
- Energy is released to the surroundings.
- $\Delta H = H(\text{products}) - H(\text{reactants}) < 0$.

Endothermic reaction



- Products are at higher energy than reactants.
- Energy is absorbed from the surroundings.
- $\Delta H = H(\text{products}) - H(\text{reactants}) > 0$.

Effect of a catalyst



- A catalyst provides an alternative reaction pathway with a lower activation energy.
- Lower E_a means a greater fraction of collisions can lead to reaction, so the reaction rate increases.
- A catalyst does not change the energy levels of reactants or products and therefore does not change ΔH .

Q8 - OR: Formulae and Mole Calculations

1. Ionic compound from Ca^{2+} and Cl^-

One Ca^{2+} ion requires two Cl^- ions for electrical neutrality, giving CaCl_2 .



Name: calcium chloride.

2. Molecular substance from one carbon joined to two oxygen atoms

The formula is CO_2 and the name is carbon dioxide.

3. Moles in 11.1 g CaCl_2

$$M(\text{CaCl}_2) = 40 + 2(35.5) = 111 \text{ g mol}^{-1}$$

$$n = \text{mass} / \text{molar mass} = 11.1 / 111 = 0.100 \text{ mol}$$

4. Moles in 4.4 g CO_2

$$M(\text{CO}_2) = 12 + 2(16) = 44 \text{ g mol}^{-1}$$

$$n = 4.4 / 44 = 0.100 \text{ mol}$$

Substance	Formula	Molar mass	Given mass	Moles
Calcium chloride	CaCl_2	111 g mol^{-1}	11.1 g	0.100 mol
Carbon dioxide	CO_2	44 g mol^{-1}	4.4 g	0.100 mol

Final Revision Checklist

- Atomic structure: atomic number, mass number, isotopes, electrons/protons/neutrons.
- Periodic trends: Group VII/17 reactivity, physical state, density and atomic size.
- Ionic bonding: electron transfer, lattice, melting point and conductivity.
- Redox: oxidation/reduction by electron transfer and oxidation number.
- Acids: reactions with metals, bases/alkalis and carbonates; ionic equations.
- Energy changes: exothermic/endothermic, activation energy, ΔH and catalysts.
- Mole calculations: particles $\leftrightarrow$ moles $\leftrightarrow$ mass, and molar mass.
- Organic chemistry: homologous series, alkanes, isomers, functional groups and ethane reactions.
- Environment and biochemistry: air pollutants, carbon monoxide, enzymes and biomolecules.

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Best wishes for your FBISE Chemistry preparation!